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EXPERIMENT NO – 6

Aim: Implementation of Prediction Algorithm (Linear Regression) using Python programming

Introduction: Linear Regression is one of the simplest and most widely used algorithms in data analytics and machine learning. It is used to find the relationship between two variables: an independent variable (input) and a dependent variable (output). The algorithm fits a straight line through the data points to model this relationship. Once the model is trained, it can be used to predict future values. In this experiment, Linear Regression is implemented in Python to predict the price of a house based on its size.

Procedure:

1. Import the required Python libraries such as **NumPy, Matplotlib, and Scikit-learn**.
2. Create a dataset containing house sizes and their corresponding prices.
3. Create a Linear Regression model using the `LinearRegression()` function.
4. Train the model using the dataset with the `fit()` method.
5. Take the house size as input from the user.
6. Use the trained model to predict the house price.
7. Display the predicted value and plot the regression line using a graph.

Program Codes:

```
# Linear Regression Prediction Example (Dynamic Input)
```

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
```

```
# Step 1: Sample dataset
```

```
# House sizes in square feet
```

```
house_size = np.array([500, 800, 1000, 1200, 1500]).reshape(-1, 1)
```

```

# House prices in thousands
house_price = np.array([150, 200, 250, 300, 350])

# Step 2: Create and train the model
model = LinearRegression()
model.fit(house_size, house_price)

# Step 3: Take input from the user
user_size = float(input("Enter the house size in sq ft to predict price: "))

# Convert input into required format
new_house_size = np.array([[user_size]])

# Step 4: Predict the price
predicted_price = model.predict(new_house_size)

# Step 5: Display the result
print("Predicted house price:", predicted_price[0], "thousand")

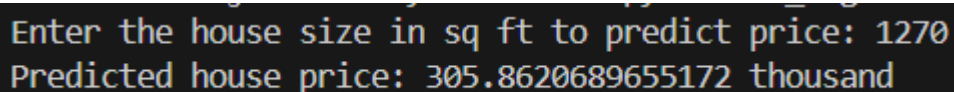
# Step 6: Plot the data and regression line
plt.scatter(house_size, house_price, label="Actual Data")
plt.plot(house_size, model.predict(house_size), label="Regression Line")

# Mark the predicted point
plt.scatter(user_size, predicted_price, label="Predicted Value")

plt.xlabel("House Size (sq ft)")
plt.ylabel("House Price (thousand)")
plt.title("Linear Regression Prediction")
plt.legend()
plt.show()

```

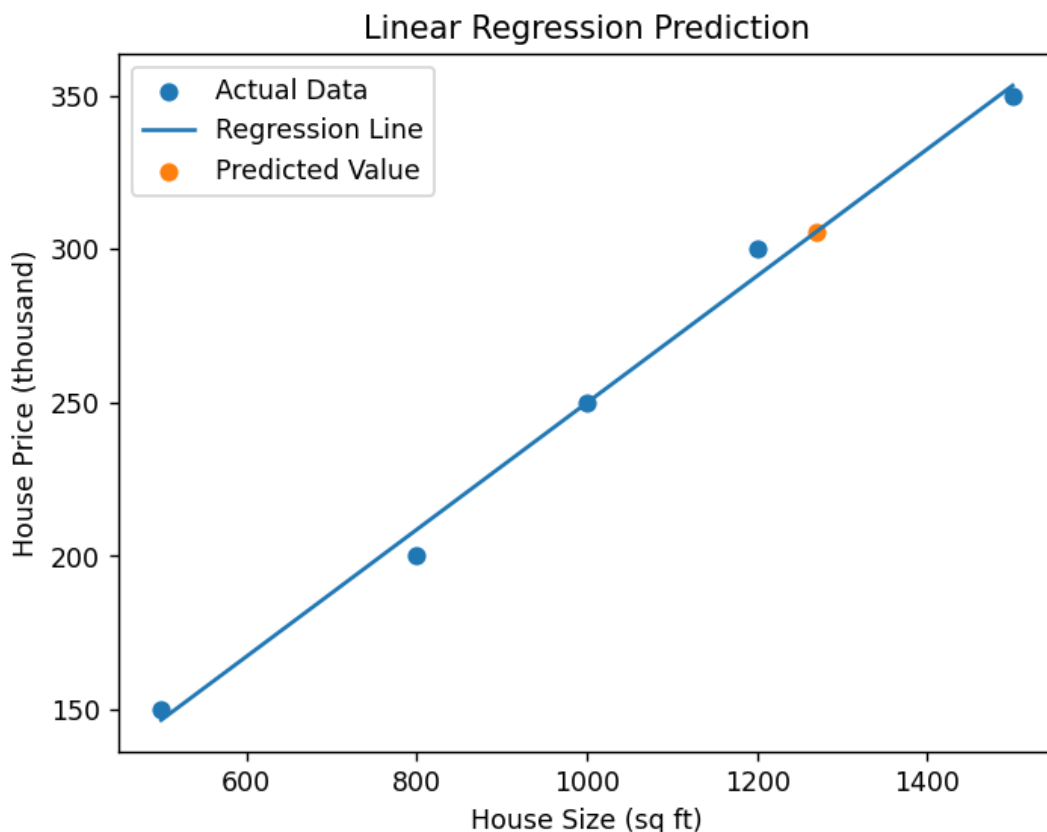
Implementation/Output snap shot:



```

Enter the house size in sq ft to predict price: 1270
Predicted house price: 305.8620689655172 thousand

```



Conclusion: In this experiment, the Linear Regression algorithm was successfully implemented using Python. The model was trained using sample data and used to predict house prices based on house size. The result demonstrates how Linear Regression can be applied for prediction in data analytics. The graphical representation also helps visualize the relationship between variables.

Review Questions:

1. Define Linear Regression. How is it used for prediction in data analytics?

Ans:

Linear Regression is a statistical and machine learning technique used to model the relationship between a dependent variable and one or more independent variables. It works by fitting a straight line to the data points. In data analytics, it is used to predict future values based on historical data.

2. Write the general equation of a simple Linear Regression model and identify its components.

Ans:

The general equation of a simple Linear Regression model is:

$$Y = \beta_0 + \beta_1 X$$

Where:

- **Y** = Predicted value (dependent variable)
- **X** = Independent variable (input feature)
- **β_0** = Intercept (value of Y when X = 0)
- **β_1** = Slope or regression coefficient (change in Y for a unit change in X)

3. What is the role of regression coefficients (β_0 and β_1) in the Linear Regression prediction model

Ans:

- **β_0 (Intercept):** Represents the starting value of the dependent variable when the independent variable is zero.
- **β_1 (Slope):** Indicates how much the dependent variable changes for each unit increase in the independent variable.

These coefficients determine the position and slope of the regression line used for prediction.

4. List any two steps involved in implementing a Linear Regression algorithm for prediction.

Ans:

1. Preparing the dataset and training the model using the fit() function.
2. Using the trained model to predict new values with the predict() function.